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An effective all-speed Riemann solver with self-similar internal structure for Euler system

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ABSTRACT

The HLLI Riemann solver proposed by Balsara is a universal scheme which is capable of resolving full family of intermediate waves with minimal dissipation for hyperbolic conservation law. However, asymptotic analyses demonstrate that such a scheme is incapable of obtaining physical solutions at low speeds due to excessive numerical dissipations of the velocity-difference terms in the momentum equations. To accommodate this deficiency, we employ the construction of the HLLEM scheme to separate shear waves which contain velocity difference in the tangential direction from contact waves and modify the quantity of the velocity difference. In addition, a function κ is adopted to damp the shear waves in the vicinity of the shockwaves to avoid shock instability phenomena. With these improvements, proper dissipation can be obtained at all speeds without suffering from the appearance of shock anomalies. Finally, a series of numerical experiments illustrate that the novel HLLIM scheme we propose is effective and robust at all speeds. Therefore, analyses in this study could provide reference for the improvement of the HLLI solver for the other hyperbolic conservation law in cases the eigenvalues vary sharply.

1. Introduction

Since proposed, Riemann solvers have become important building blocks of modern numerical methods for solving Euler and other conservation systems [1]. Also, a great many approximate Riemann solvers, such as the Godunov scheme [2], the Roe scheme [3], the Osher scheme [4], the AUSM-type schemes [5–7] and the HLL-type schemes [8–10], etc., have been extensively used in the design of the aircraft.

Among these methods, the HLL-type schemes satisfy the positivity condition without artificial parameters. Also, it is not necessary to know the entire eigenstructure of the flux Jacobian as the Roe or AUSM-type methods. Therefore, it can be easily extended to other large hyperbolic conservation laws, such as the multiphase flow models [11,12]. As the first version HLL-type scheme, the original HLL method is designed to handle shocks and expansion fans. However, it is based on the two-wave model and cannot resolve linearly degenerate waves accurately. In order to improve such defect, the HLLC scheme [13], the HLLEM scheme [14] and the HLLC-SWM scheme [15] etc. are proposed. Numerical tests indicate that they are capable of sharply resolving linear contact discontinuities and shear layers. In addition, inspired by the self-similar property of the hyperbolic conservative Euler equations, Balsara et al. introduced a derivative paradigm based on the self-similar variables and developed a complete Riemann solver called HLLI [10,16].

In spite of the success achieved, all these HLL-type Riemann solvers suffer from the unphysical shock instability phenomenon in capturing strong shocks. For example, the HLLI method modifies its dissipation term according to the construction of the Roe scheme, which may cause inappropriate numerical viscosity for rarefaction as original Roe scheme. Therefore, the robustness and the accuracy of the dissipation term should be analyzed in details, and a good Riemann solver should not only capture all nonlinear and linear waves accurately, but be able to overcome unphysical behaviors at certain conditions as well.

Furthermore, the HLL-type scheme will face two major deficiencies for incompressible or low-speed flows: non-physical behavior and difficult convergence [17,18]. These problems are usually embedded in high speed flows, such as the boundary layer and the shear layer [19]. A general way to overcome this problem is to use preconditioning technology [20,21]. However, when simulating mixed flows with low and high *Mach* numbers, this problem may not be well-resolved due to the global cut-off problem [22].

In this study, we conduct analyses on the HLLI scheme for both high-speed and low-speed flows. Shock instability and inaccurate pressure

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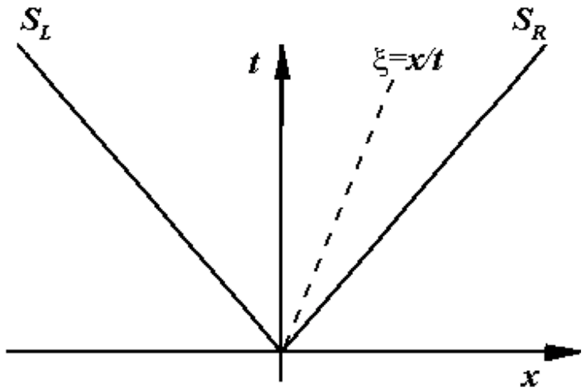


Fig. 1. Sketch of HLL approximate Riemann solver.

scales are found for the HLLI scheme. Then, by introducing two switch functions to control the dissipation both in the vicinity of shockwaves and at low speeds, the HLLIM scheme is proposed to overcome these unphysical solutions in both cases. In addition, theoretical and numerical analyses are performed to confirm the effect of such technique.

The rest of the paper is organized as follows. In Section 2, we will briefly introduce the compressible Euler equations Section 3. reviews the construction of the HLLI scheme. Then we perform a matrix stability analysis and an asymptotic analysis in Section 4. Moreover, the HLLIM scheme is proposed based on these analyses. To validate the characteristics of the proposed scheme, systematic numerical experiments are presented in Section 5. The last section gives concluding remarks.

2. Euler equations

Euler equations are usually used to describe the motion of inviscid flows which can be written in the following differential form [23]:

$$\frac{\partial \mathbf{q}}{\partial t} + \frac{\partial \mathbf{f}}{\partial \mathbf{x}} = \mathbf{0} \tag{1}$$

where $\mathbf{q}$ is the conservative variable vector and $\mathbf{f}$ denotes the inviscid flux vector.

For two-dimensional cases, we have

$$\boldsymbol{q} = (\rho, \rho u, \rho v, \rho e)^T \tag{2}$$

$$\mathbf{f} = (f_1, f_2), \quad \mathbf{x} = (x_1, x_2) \tag{3}$$

with

$$\begin{aligned} f_1 &= \left(\rho, \rho u^2 + p, \rho u v, u(\rho e + p)\right)^T \\ f_2 &= \left(\rho, \rho u v, \rho v^2 + p, v(\rho e + p)\right)^T \end{aligned} \tag{4}$$

where ρ is density, u and v are velocity components in x_1, x_2 direction, respectively. In addition, p is static pressure and T is total specific energy where ε denotes the specific internal energy. Furthermore, perfect gas law ($p = (\gamma - 1)\rho\varepsilon$) is adopted to close the equations with the ratio of specific heats of $\gamma = 1.4$.

3. Recap of the construction of the HLLI scheme

The HLLI Riemann solver [10,16], designed based on the self-similarity property, is proposed by Balsara et al. who consider the intermediated wave as its eigenstructure for one dimensional Euler system. Hence the HLLI solver is a complete Riemann solver capable of resolving contact discontinuities or shear waves and is easy to extend to multidimensional Riemann solvers. The derivation process of HLLI scheme will be given briefly in this section.

Based on a classical similarity variable $\xi = x/t$, the conservative variable vector and flux can be rewritten as $\mathbf{q}(\xi)$ and $\mathbf{f}(\xi)$, both of which contain only the similar variable. Taking into account of the intermediate wave of the Euler system, as proposed in [10], the approximate Riemann solution is given as:

$$\mathbf{q}(\xi) = \begin{cases} \mathbf{q}_L, & \text{if } \xi \le S_L \\ \mathbf{q}_* + \mathbf{R}_*(2\boldsymbol{\delta}_*)\mathbf{L}_* \dfrac{\mathbf{q}_R - \mathbf{q}_L}{\Delta\xi}(\xi - \xi_c) & \text{if } S_L \le \xi \le S_R \\ \mathbf{q}_R, & \text{if } \xi \ge S_R \end{cases} \tag{5}$$

Here, $\mathbf{R}_*, \mathbf{L}_*$ are the right and left eigenvector matrix of the linearly degenerate intermediate characteristics fields and $\boldsymbol{\delta}_*$ is the associated diagonal matrix.

For simplification, we use the normalized similarity variable based on the wave structure of HLL-type scheme (Fig. 1):

$$\widehat{\xi} = \frac{\xi - \xi_c}{\Delta\xi}, \quad \text{with} \quad \xi_c = \frac{S_R + S_L}{2}, \quad \Delta\xi = S_R - S_L \tag{6}$$

where S_L, S_R are the left and right moving wave speeds, respectively. In this paper, we adopt the following estimation as [8]:

$$S_L = \min(0, \Lambda(\mathbf{q}_L), \Lambda(\widetilde{\mathbf{q}})), S_R = \max(0, \Lambda(\mathbf{q}_R), \Lambda(\widetilde{\mathbf{q}})) \tag{7}$$

where $\Lambda(\mathbf{q})$ is the diagonal matrix of the Jacobian matrix $\mathbf{A}(\mathbf{q}) = \frac{\partial \mathbf{f}}{\partial \mathbf{q}}$ and $\widetilde{\ }$ denotes the Roe-average quantities at the interface.

Based on the normalized similarity variable, the conservative variable and flux can be expanded as:

$$\mathbf{q}(\widehat{\xi}) = \overline{\mathbf{q}} + \mathbf{q}_{\widehat{\xi}}\widehat{\xi} \quad \text{and} \quad \mathbf{f}(\widehat{\xi}) = \overline{\mathbf{f}} + \mathbf{f}_{\widehat{\xi}}\widehat{\xi} = \overline{\mathbf{f}} + \frac{\partial \overline{\mathbf{f}}}{\partial \overline{\mathbf{q}}}\frac{\partial \mathbf{q}}{\partial \widehat{\xi}}\widehat{\xi} = \overline{\mathbf{f}} + \overline{\mathbf{A}}\mathbf{q}_{\widehat{\xi}}\widehat{\xi} \tag{8}$$

Here $\overline{\ }$ denotes the cell average process and the flux and state variables can be evaluated using approximate Riemann solver. Also, the linear approximation solution (Eq. (5)) can be simplified in the following form:

$$\mathbf{q}(\widetilde{\xi}) = \begin{cases} \mathbf{q}_L, & \text{if } \widetilde{\xi} \le -\dfrac{1}{2} \\ \mathbf{q}_* + \mathbf{R}_*(2\boldsymbol{\delta}_*)\mathbf{L}_*(\mathbf{q}_R - \mathbf{q}_L)\widetilde{\xi} & \text{if } -\dfrac{1}{2} \le \widetilde{\xi} \le \dfrac{1}{2} \\ \mathbf{q}_R, & \text{if } \widetilde{\xi} \ge \dfrac{1}{2} \end{cases} \tag{9}$$

And the Euler system is rewritten as:

$$\frac{1}{\Delta\xi}\frac{\partial(\mathbf{f} - (\xi_c + \widehat{\xi}\Delta\xi)\mathbf{q})}{\partial\xi} + \mathbf{q} = 0 \tag{10}$$

Multiplying a test function $v(\widehat{\xi})$ on both sides of Eq. (10) and using the formula of integration by parts, we obtain the expression:

$$\frac{1}{\Delta\xi}\frac{\partial(v(\widehat{\xi})(\mathbf{f} - (\xi_c + \widehat{\xi}\Delta\xi)\mathbf{q}))}{\partial\widehat{\xi}} - \frac{1}{\Delta\xi}(\mathbf{f} - (\xi_c + \widehat{\xi}\Delta\xi)\mathbf{q})\frac{\partial v(\widehat{\xi})}{\partial\widehat{\xi}} + v(\widehat{\xi})\mathbf{q} = 0 \tag{11}$$

Then we do Galerkin projection with different test functions to obtain the expression of the averaged flux in the star region($\xi \in [S_L, S_R]$).

We first adopt a test function of $v(\widehat{\xi}) = 1$ and integrate Eq. (11) over the region $\xi \in [S_L, S_R]$. The integral average of the classical HLL state vector is achieved:

$$\overline{\mathbf{q}}_{HLL} = \frac{\mathbf{f}_L - \mathbf{f}_R + S_R\mathbf{q}_R - S_L\mathbf{q}_L}{\Delta\xi} \tag{12}$$

Then we use another test function $v(\widehat{\xi}) = \widehat{\xi}$ and acquire a different form:

$$\overline{\mathbf{f}} = \frac{\Delta\xi}{6}\mathbf{q}_{\widehat{\xi}} + \frac{\mathbf{f}_L + \mathbf{f}_R - S_R\mathbf{q}_R - S_L\mathbf{q}_L}{2} + \xi_c\overline{\mathbf{q}}_{HLL} \xrightarrow{\text{implement } \overline{\mathbf{q}}_{HLL}} \frac{\Delta\xi}{6}\mathbf{q}_{\widehat{\xi}} + \overline{\mathbf{f}}_{HLL} \tag{13}$$

where $\bar{\mathbf{f}}_{HLL} = \frac{S_R\mathbf{f}_L - S_L\mathbf{f}_R + S_RS_L(\mathbf{q}_R-\mathbf{q}_L)}{\Delta\xi}$, which is the classical HLL flux expression.

On account of the complete Riemann solution structure in Eq. (9), we have $\mathbf{q}_{\widehat{\xi}} = \mathbf{R}_*(2\boldsymbol{\delta}_*)\mathbf{L}_*(\mathbf{q}_R - \mathbf{q}_L)$. Combining Eq. (8), we can obtain the numerical flux at the cell interface($\xi = 0$ *i.e.* $\widehat{\xi} = \frac{-\xi_c}{\Delta\xi}$):

$$\begin{aligned} &\mathbf{f}_{HLLI} = \bar{\mathbf{f}} + \bar{\mathbf{A}}\mathbf{q}_{\widehat{\xi}}\widehat{\xi} = \left(\frac{\Delta\xi}{6}\mathbf{q}_{\widehat{\xi}} + \bar{\mathbf{f}}_{HLL}\right) - \frac{\xi_c}{\Delta\xi}\bar{\mathbf{A}}\mathbf{q}_{\widehat{\xi}} \\ &\textit{where for the linear degenerate wave}: \\ &\bar{\mathbf{A}}\mathbf{q}_{\widehat{\xi}} = \mathbf{R}_*\boldsymbol{\Lambda}_*\mathbf{L}_*\mathbf{R}_*(2\boldsymbol{\delta}_*)\mathbf{L}_*(\mathbf{q}_R-\mathbf{q}_L) = \mathbf{R}_*\boldsymbol{\Lambda}_*(2\boldsymbol{\delta}_*)\mathbf{L}_*(\mathbf{q}_R-\mathbf{q}_L) \\ &so\ \mathbf{f}_{HLLI} = \bar{\mathbf{f}}_{HLL} - \frac{1}{2}\mathbf{R}_*\left(-\frac{\Delta\xi}{3}(2\boldsymbol{\delta}_*) + \frac{2\xi_c}{\Delta\xi}\boldsymbol{\Lambda}_*(2\boldsymbol{\delta}_*)\right)\mathbf{L}_*(\mathbf{q}_R-\mathbf{q}_L) \end{aligned} \tag{14}$$

Above is the formulation of the HLLI scheme. In addition, the choice of the value of $\boldsymbol{\delta}_*$ is carefully designed for the consideration of both accuracy and stability. In this paper we follow the choice of Balsara [16]:

$$\delta_i = \begin{cases} -\dfrac{3S_RS_L}{(S_R-S_L)^2} & when\ (S_R-S_L)^2 \Big/ 3-\lambda_i(S_R+S_L)\le 0 \\ \min\left(\dfrac{S_R\lambda_i^- + S_L\lambda_i^+ - S_RS_L}{(S_R-S_L)^2/3-\lambda_i(S_R+S_L)}, -\dfrac{3S_RS_L}{(S_R-S_L)^2}\right) & otherwise \end{cases} \tag{15}$$

where subscript i indicates the i_{th} corresponding wave structure, and λ indicates the eigenvalues of the Jacobian matrix. Futhermore, $\lambda_i^{\pm} = \frac{1}{2}(\lambda_i \pm |\lambda_i|)$. when $\lambda_i \to 0$, $\delta_i = -\frac{3S_RS_L}{(S_R-S_L)^2}$. Substituting this into Eq. (14), we can find that the numerical dissipation is zero which guarantees the resolving ability of stationary contact discontinuities.

4. An improved HLLI method (HLLIM) for all-speed flows

4.1. Matrix stability analysis

In this section, we investigate the mechanism of the shock instability problem based on a matrix stability method proposed initially by Dumbser [24]. The analysis studies the rate of change of the perturbations and we briefly introduce the procedure here. Calculations are performed on a two dimensional computational region with regular Cartesian cells. The total grid number is $M = 11 \times 11$ for the present study. In the initial we suppose an isolated steady shock wave with perturbations $\mathbf{q}_{i,j} = \mathbf{q}_{i,j}^0 + \delta\mathbf{q}_{i,j}$ where $\delta\mathbf{q}_{i,j}$ denotes a small perturbation vector with a magnitude of 10^{-4}. Substituting the initial condition into a finite volume formulation, we obtain the discrete form:

$$\frac{d\mathbf{q}_{i,j}}{dt} = -\frac{\sum_{k=1}^{4}\mathbf{f}_{i,j}A_{i,j}}{\Omega} \tag{16}$$

where $\mathbf{f}_{i,j}$ is the numerical flux function, $A_{i,j}$ is the area of the interface, and Ω is the volume of cell i,j. It should be mentioned that only first-order reconstruction method are taken into account for this analysis. After taking several evolutions, we can finally get the rate of change of the initial perturbation in a form of linear ODE(ordinary differential equation) as:

$$\frac{d}{dt}\begin{pmatrix}\delta\mathbf{q}_1\\ \vdots \\ \delta\mathbf{q}_M\end{pmatrix} = S\begin{pmatrix}\delta\mathbf{q}_1\\ \vdots \\ \delta\mathbf{q}_M\end{pmatrix} \tag{17}$$

where S is the stability matrix determined by the initial conditions and numerical flux method. A scheme is termed stable if the maximum of the real part of the eigenvalues (Max($Re(\theta)$)) of S matrix is non-positive according to this linear perturbation analysis.

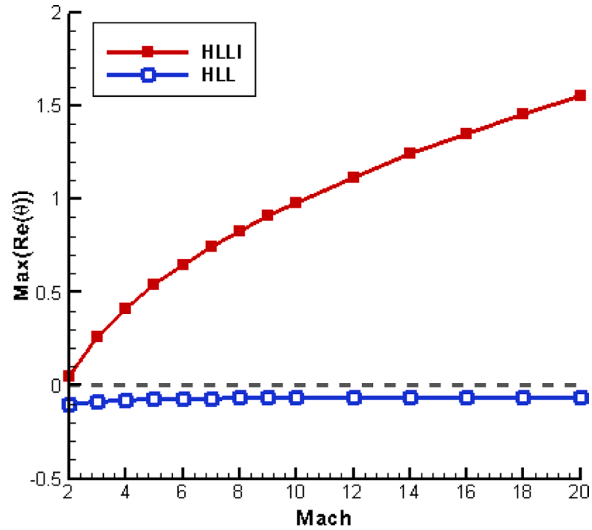


Fig. 2. Maximal real part of eigenvalues of stability matrix VS. *Mach* number.

For the present study, we consider a normal steady shock with different *Mach* number. The upstream and downstream states are given by the Rankine-Hugoniot conditions:

$$\begin{aligned} \mathbf{q}_L &= \left(1, 1, 0, \left(\gamma(\gamma-1)M_0{}^2\right)^{-1} + 1\Big/2\right)^T, \qquad x < 0.5 \\ \mathbf{q}_R &= \left(f(M_0), 1, 0, g(M_0)\left(\gamma(\gamma-1)M_0{}^2\right)^{-1} + (2f(M_0))^{-1}\right)^T, \qquad x > 0.5 \end{aligned} \tag{18}$$

where M_0 is the upstream *Mach* number and

$$f(M_0) = \left(\frac{2}{(\gamma+1)M_0{}^2} + \frac{\gamma-1}{\gamma+1}\right)^{-1}, \quad g(M_0) = \frac{2\gamma M_0{}^2}{\gamma+1} - \frac{\gamma-1}{\gamma+1} \tag{19}$$

The top and bottom boundaries are set as periodic boundary conditions. For the left and right boundary, post and pre shock condition are implemented respectively.Fig. 2 presents the results of HLLI and HLL schemes for different *Mach* numbers. It can be seen that the HLLI scheme is unstable when the *Mach* number is above 2, and the instability is becoming more severe with the increase of the *Mach* number. HLL scheme, however, is stable for a wide range of *Mach* number.

4.2. Asymptotic analysis

In this section, we perform an asymptotic analysis according to the Refs [25,26] to study the low-speed properties of the HLLI scheme.

For simplicity, we consider a two dimensional cartesian grid with uniformed size $\widetilde{\delta}$. The notation $\nu(\mathbf{i}) = \{(i\pm1, j), (i, j\pm1)\}$ denotes the neighbouring grids for the grid node$\mathbf{i} = (i,j)$. Each grid is with a control volume of $C_{\mathbf{i}} = [(i-1/2)\widetilde{\delta}, (i+1/2)\widetilde{\delta}] \times [(j-1/2)\widetilde{\delta}, (j+1/2)\widetilde{\delta}]$. In addition, we denote $\vec{\mathbf{n}}_{\mathbf{il}} = [(n_x)_{\mathbf{il}}, (n_y)_{\mathbf{il}}]^{\mathrm{T}}$ as unit outer normal vector from grid node $\mathbf{i}$ to grid node $\mathbf{l}$ and $\mathbf{il}$ is the index for the edge between grid node $\mathbf{i}$ and grid node $\mathbf{l}$.

According the definition in Eqs. (7) and (15), we can obtain that the wave speed and δ function can be approximated as follows when the *Mach* number goes to zero:

$$S_L \approx -\widetilde{c} \tag{20}$$

$$S_R \approx \widetilde{c} \tag{21}$$

$$\delta_i \approx \frac{3}{4} \tag{22}$$

Hence we have the approximation according to Eq. (6)

$$\Delta\xi \approx 2\widetilde{c}, \quad \xi_c \approx 0 \tag{23}$$

Also, the eigenvalues of Jacobian matrix is approximated as

$$\lambda_i \approx 0, 0, \widetilde{c}, -\widetilde{c} \tag{24}$$

On the other hand, for a Euler system in curvilinear form, we have the intermediate degenerate linear characteristic field.:

$$\mathbf{R}_* = \begin{pmatrix} 1, u, v, \left(u^2+v^2\right)/2 \\ 0, -n_y, -n_x, vn_x - un_y \end{pmatrix}^T, \quad \boldsymbol{\Lambda}_* = \begin{pmatrix} un_x + vn_y, 0 \\ 0, un_x + vn_y \end{pmatrix}, \quad \mathbf{L}_*(\mathbf{q}_R - \mathbf{q}_L) = \begin{pmatrix} -\left(\Delta p - c^2\Delta\rho\right)/c^2 \\ \rho\Delta\left(vn_x - un_y\right) \end{pmatrix} \tag{25}$$

where $\Delta(\bullet)$ denotes $(\bullet)_R - (\bullet)_L$.

Based on the above relationships, the second term $-\frac{1}{2}\mathbf{R}_*\left(-\frac{\Delta\xi}{3}(2\delta) + \frac{2\xi_c}{\Delta\xi}\boldsymbol{\Lambda}_*(2\delta)\right)\mathbf{L}_*(\mathbf{q}_R - \mathbf{q}_L)$ of HLLI flux (Eq. (14)) can be simplified to $\frac{\widetilde{c}}{2}\mathbf{R}_*\mathbf{L}_*(\mathbf{q}_R - \mathbf{q}_L)$. Therefore, the numerical flux (Eq. (14)) can be approximated as:

$$\mathbf{f}_{numerical} = \frac{1}{2}(\mathbf{f}_L + \mathbf{f}_R) - \frac{\widetilde{c}}{2}\begin{pmatrix} \Delta\rho \\ \Delta(\rho u) \\ \Delta(\rho v) \\ \Delta(\rho e) \end{pmatrix} + \frac{\widetilde{c}}{2}\begin{pmatrix} -\left(\Delta p - c^2\Delta\rho\right)/c^2 \\ -u\left(\Delta p - c^2\Delta\rho\right)/c^2 - n_y\rho\Delta\left(vn_x - un_y\right) \\ -v\left(\Delta p - c^2\Delta\rho\right)/c^2 - n_x\rho\Delta\left(vn_x - un_y\right) \\ \left(u^2+v^2\right)\left(\Delta p - c^2\Delta\rho\right)/2c^2 + \rho\left(vn_x - un_y\right)\Delta\left(vn_x - un_y\right) \end{pmatrix} \tag{26}$$

By integrating Euler equations with a first-order finite volume method using the HLLI scheme, we can obtain the following semi-discrete equations:

Continuity equation:

$$\widetilde{\delta}\frac{\partial\rho_i}{\partial t} + \frac{1}{2}\sum_{l\in v(i)}\rho_l\mathbf{u}_l\cdot\vec{\mathbf{n}}_{il} - \frac{1}{2c_{il}}\sum_{l\in v(i)}\Delta_{il}p = 0 \tag{27}$$

X-momentum equation:

$$\widetilde{\delta}\frac{\partial\rho_i u_i}{\partial t} + \frac{1}{2}\sum_{l\in v(i)}\left[\rho_l u_l\mathbf{u}_l\cdot\vec{\mathbf{n}}_{il} + p_l(n_x)_{il}\right] + \frac{1}{2c_{il}}\sum_{l\in v(i)}u_{il}\Delta_{il}p - \frac{c_{il}}{2}\sum_{l\in v(i)}\left[\Delta_{il}(\rho u) - u_{il}\Delta_{il}\rho + \left(n_y\right)_{il}(n_x)_{il}\rho_{il}\Delta_{il}v - \left(n_y\right)_{il}^2\rho_{il}\Delta_{il}u\right] = 0 \tag{28}$$

Y-momentum equation:

$$\widetilde{\delta}\frac{\partial\rho_i v_i}{\partial t} + \frac{1}{2}\sum_{l\in v(i)}\left[\rho_l v_l\mathbf{u}_l\cdot\vec{\mathbf{n}}_{il} + p_l\left(n_y\right)_{il}\right] + \frac{1}{2c_{il}}\sum_{l\in v(i)}v_{il}\Delta_{il}p - \frac{c_{il}}{2}\sum_{l\in v(i)}\left[\Delta_{il}(\rho v) - v_{il}\Delta_{il}\rho + (n_x)_{il}^2\rho_{il}\Delta_{il}v - (n_x)_{il}\left(n_y\right)_{il}\rho_{il}\Delta_{il}u\right] = 0 \tag{29}$$

Energy equation:

$$\widetilde{\delta}\frac{\partial\rho_i e_i}{\partial t} + \frac{1}{2}\sum_{l\in v(i)}\left[(\rho_l e_l + p_l)\mathbf{u}_l\cdot\vec{\mathbf{n}}_{il}\right] + \frac{1}{4c_{il}}\sum_{l\in v(i)}\left(u_{il}^2 + v_{il}^2\right)\Delta_{il}p - \frac{c_{il}}{2}\sum_{l\in v(i)}\left[\Delta_{il}(\rho e) - \frac{\left(u_{il}^2 + v_{il}^2\right)\Delta_{il}\rho}{2} + \rho_{il}\left((n_x)_{il}v_{il} - \left(n_y\right)_{il}u_{il}\right)\left((n_x)_{il}\Delta_{il}v - \left(n_y\right)_{il}\Delta_{il}u\right)\right] = 0 \tag{30}$$

To simplify the process of the asympototic analysis, we normalized the physical variables with respect to parameters: $\rho^* = \max(\rho), u^* = \max(\sqrt{u^2+v^2}), c^* = \sqrt{\gamma\max(p)/\rho^*}$. Then the following non-dimensional flow variables are obtained:

$$\widehat{\rho} = \frac{\rho}{\rho^*}, \widehat{u} = \frac{u}{u^*}, \widehat{v} = \frac{v}{v^*}, \widehat{x} = \frac{x}{x^*}, \widehat{y} = \frac{y}{x^*}, \quad \widehat{p} = \frac{p}{\rho^*(c^*)^2}, \widehat{e} = \frac{e}{(c^*)^2}, \widehat{t} = \frac{tu^*}{x^*} \tag{31}$$

where x^* is an arbitrary length scale.

Then the Eqs. (19)-(22) can be expressed with the normalized variables:

Continuity equation:

$$\widetilde{\delta}\frac{\partial\widehat{\rho}_i}{\partial\widehat{t}} + \frac{1}{2}\sum_{l\in v(i)}\widehat{\rho}_l\widehat{\mathbf{u}}_l\cdot\vec{\mathbf{n}}_{il} - \frac{1}{\mathrm{Ma}}\frac{1}{2\widehat{c}_{il}}\sum_{l\in v(i)}\Delta_{il}\widehat{p} = 0 \tag{32}$$

X-momentum equation:

$$\widetilde{\delta}\frac{\partial\widehat{\rho}_i\widehat{u}_i}{\partial\widehat{t}} + \frac{1}{2}\sum_{l\in v(i)}\widehat{\rho}_l\widehat{u}_l\widehat{\mathbf{u}}_l\cdot\vec{\mathbf{n}}_{il} + \frac{1}{2\mathrm{Ma}^2}\sum_{l\in v(i)}\widehat{p}_l(n_x)_{il} + \frac{1}{\mathrm{Ma}}\frac{1}{2\widehat{c}_{il}}\sum_{l\in v(i)}\widehat{u}_{il}\Delta_{il}\widehat{p} - \frac{1}{\mathrm{Ma}}\frac{\widehat{c}_{il}}{2}\sum_{l\in v(i)}\left[\Delta_{il}(\widehat{\rho}\widehat{u}) - \widehat{u}_{il}\Delta_{il}\widehat{\rho} + \left(n_y\right)_{il}(n_x)_{il}\widehat{\rho}_{il}\Delta_{il}\widehat{v} - \left(n_y\right)_{il}^2\widehat{\rho}_{il}\Delta_{il}\widehat{u}\right] = 0 \tag{33}$$

Y-momentum equation:

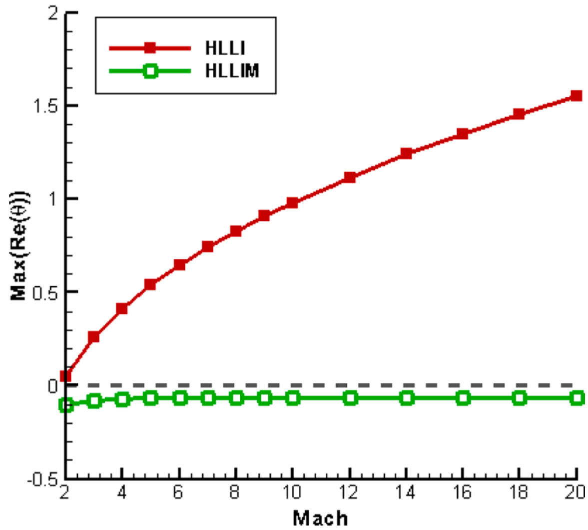


Fig. 3. Comparison of maximal real part of eigenvalues of stability matrix VS. *Mach* number.

$$\widehat{\delta}\frac{\partial\widehat{\rho}_{\mathbf{i}}\widehat{v}_{\mathbf{i}}}{\partial\widehat{t}}+\frac{1}{2}\sum_{\mathbf{l}\in v(\mathbf{i})}\widehat{\rho}_{\mathbf{l}}\widehat{v}_{\mathbf{l}}\widehat{\mathbf{u}}_{\mathbf{l}}\cdot\overrightarrow{\mathbf{n}}_{\mathbf{il}}+\frac{1}{2\mathrm{Ma}^2}\sum_{\mathbf{l}\in v(\mathbf{i})}\widehat{p}_{\mathbf{l}}\left(n_y\right)_{\mathbf{il}}+\frac{1}{\mathrm{Ma}}\frac{1}{2\widehat{c}_{\mathbf{il}}}\sum_{\mathbf{l}\in v(\mathbf{i})}\widehat{v}_{\mathbf{il}}\Delta_{\mathbf{il}}\widehat{p}$$
$$-\frac{1}{\mathrm{Ma}}\frac{\widehat{c}_{\mathbf{il}}}{2}\sum_{\mathbf{l}\in v(\mathbf{i})}\left[\Delta_{\mathbf{il}}(\widehat{\rho}\widehat{v})\;-\;\widehat{v}_{\mathbf{il}}\Delta_{\mathbf{il}}\widehat{\rho}+\left(n_y\right)_{\mathbf{il}}{}^2\widehat{\rho}_{\mathbf{il}}\Delta_{\mathbf{il}}\widehat{v}-(n_x)_{\mathbf{il}}\left(n_y\right)_{\mathbf{il}}\widehat{\rho}_{\mathbf{il}}\Delta_{\mathbf{il}}\widehat{u}\right]=0 \tag{34}$$

Energy equation:

$$\widehat{\delta}\frac{\partial\widehat{\rho}_{\mathbf{i}}\widehat{e}_{\mathbf{i}}}{\partial\widehat{t}}+\frac{1}{2}\sum_{\mathbf{l}\in v(\mathbf{i})}\left[(\widehat{\rho}_{\mathbf{l}}\widehat{e}_{\mathbf{l}}+\widehat{p}_{\mathbf{l}})\widehat{\mathbf{u}}_{\mathbf{l}}\cdot\overrightarrow{\mathbf{n}}_{\mathbf{il}}\right]+\frac{\mathrm{Ma}}{4\widehat{c}_{\mathbf{il}}}\sum_{\mathbf{l}\in v(\mathbf{i})}\left(\widehat{u}_{\mathbf{il}}{}^2+\widehat{v}_{\mathbf{il}}{}^2\right)\Delta_{\mathbf{il}}p$$
$$-\frac{\widehat{c}_{\mathbf{il}}}{2}\sum_{\mathbf{l}\in v(\mathbf{i})}\left[\frac{1}{\mathrm{Ma}}\Delta_{\mathbf{il}}(\widehat{\rho}\widehat{e})\;-\;\frac{\mathrm{Ma}\left(\widehat{u}_{\mathbf{il}}{}^2+\widehat{v}_{\mathbf{il}}{}^2\right)\Delta_{\mathbf{il}}\widehat{\rho}}{2}+\mathrm{Ma}\widehat{\rho}_{\mathbf{il}}\left((n_x)_{\mathbf{il}}\widehat{v}_{\mathbf{il}}-\left(n_y\right)_{\mathbf{il}}\widehat{u}_{\mathbf{il}}\right)\left((n_x)_{\mathbf{il}}\Delta_{\mathbf{il}}\widehat{v}-\left(n_y\right)_{\mathbf{il}}\Delta_{\mathbf{il}}\widehat{u}\right)\right]=0 \tag{35}$$

where $\mathrm{Ma}=\frac{u^*}{c^*}$ is the reference *Mach* number and $\Delta_{\mathbf{il}}(\bullet)=(\bullet)_{\mathbf{i}}-(\bullet)_{\mathbf{l}}$.

As the *Mach* number is small for the condition we investigated, the physical quantities($\phi=\widehat{\rho},\widehat{p},\widehat{e},\widehat{u},\widehat{v}$) can be expand into powers of the reference *Mach* number as:

$$\phi=\phi^{(0)}+\mathrm{Ma}\phi^{(1)}+\mathrm{Ma}^2\phi^{(2)}+O\left(\mathrm{Ma}^3\right) \tag{36}$$

Combining Eqs. (32)–(35) together and collecting the terms of the same power of the *Mach* number, we can obtain the following

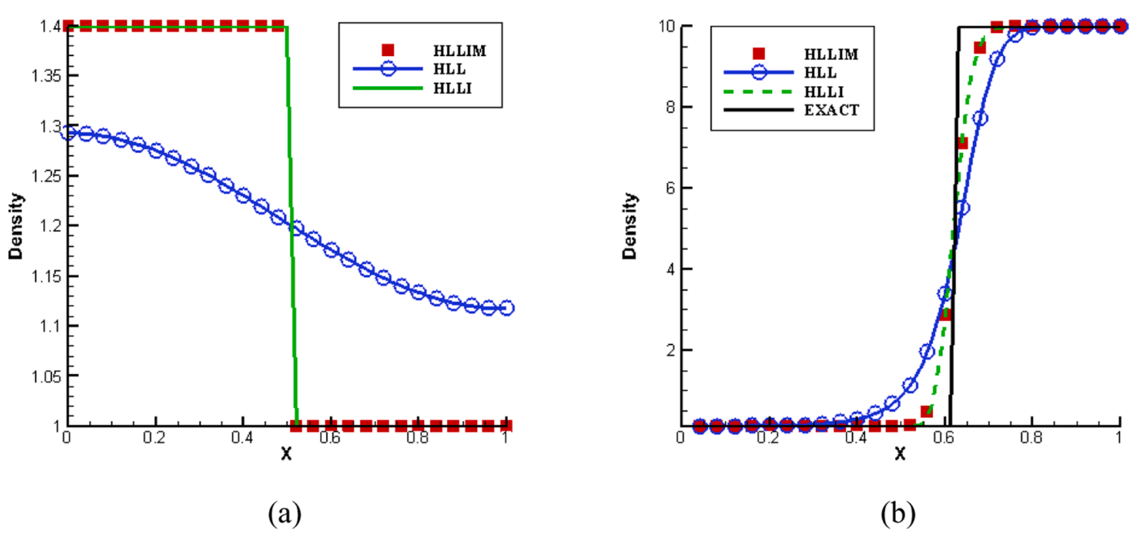
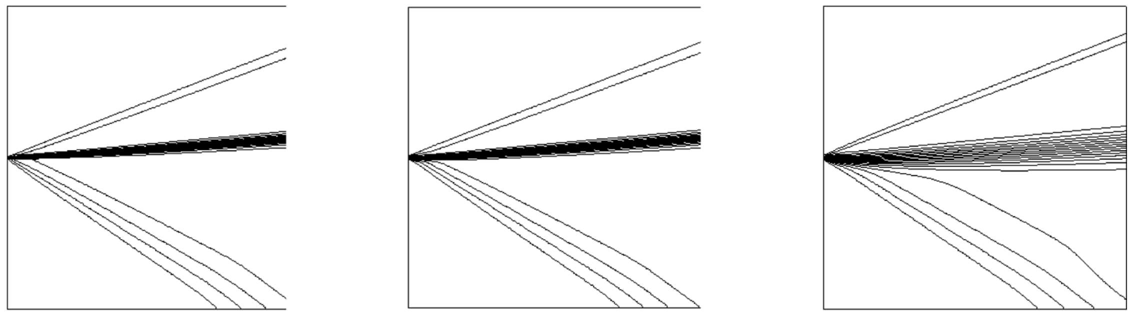


Fig. 4. Density distribution of a contact discontinuity. (a)steady contact discontinuity; (b) moving contact discontinuity.



(a) HLLI (b) HLLIM (c) :HLL

Fig. 5. Density contours for supersonic jet problem.

relationships:

- Order $1/\mathrm{Ma}^2$

$$\begin{aligned} &\widehat{p}^0_{i+1,j}-\widehat{p}^0_{i-1,j}=0\\ &\widehat{p}^0_{i,j+1}-\widehat{p}^0_{i,j-1}=0 \end{aligned} \tag{37}$$

- Order $1/\mathrm{Ma}$

(a) continuity equation:

$$\frac{1}{2\widehat{c}^0_{\mathrm{il}}}\sum_{\mathrm{l}\in v(\mathrm{i})}\Delta_{\mathrm{il}}\widehat{\rho}^0=0 \tag{38}$$

(b) momentum equation:

$$\sum_{\mathrm{l}\in v(\mathrm{i})}\left[\widehat{p}^1_{\mathrm{l}}(n_x)_{\mathrm{il}}\right]=-\frac{1}{2\widehat{c}^0_{\mathrm{il}}}\sum_{\mathrm{l}\in v(\mathrm{i})}\widehat{u}^0_{\mathrm{il}}\Delta_{\mathrm{il}}\widehat{\rho}^0$$
$$+\frac{\widehat{c}^0_{\mathrm{il}}}{2}\sum_{\mathrm{l}\in v(\mathrm{i})}\left[\Delta_{\mathrm{il}}\left(\widehat{\rho}^0\widehat{u}^0\right)-\widehat{u}^0_{\mathrm{il}}\Delta_{\mathrm{il}}\widehat{\rho}^0+(n_y)_{\mathrm{il}}(n_x)_{\mathrm{il}}\widehat{\rho}^0_{\mathrm{il}}\Delta_{\mathrm{il}}\widehat{v}^0-(n_y)_{\mathrm{il}}(n_y)_{\mathrm{il}}\widehat{\rho}^0_{\mathrm{il}}\Delta_{\mathrm{il}}\widehat{u}^0\right]\neq 0 \tag{39}$$

$$\sum_{\mathrm{l}\in v(\mathrm{i})}\left[\widehat{p}^1_{\mathrm{l}}(n_y)_{\mathrm{il}}\right]=-\frac{1}{2\widehat{c}^0_{\mathrm{il}}}\sum_{\mathrm{l}\in v(\mathrm{i})}\widehat{v}^0_{\mathrm{il}}\Delta_{\mathrm{il}}\widehat{\rho}^0$$
$$+\frac{\widehat{c}^0_{\mathrm{il}}}{2}\sum_{\mathrm{l}\in v(\mathrm{i})}\left[\Delta_{\mathrm{il}}\left(\widehat{\rho}^0\widehat{v}^0\right)-\widehat{v}^0_{\mathrm{il}}\Delta_{\mathrm{il}}\widehat{\rho}^0+(n_y)_{\mathrm{il}}(n_y)_{\mathrm{il}}\widehat{\rho}^0_{\mathrm{il}}\Delta_{\mathrm{il}}\widehat{v}^0-(n_x)_{\mathrm{il}}(n_y)_{\mathrm{il}}\widehat{\rho}^0_{\mathrm{il}}\Delta_{\mathrm{il}}\widehat{u}^0\right]\neq 0 \tag{40}$$

(c) energy equation:

$$\frac{\widehat{c}^0_{\mathrm{il}}}{2}\sum_{\mathrm{l}\in v(\mathrm{i})}\Delta_{\mathrm{il}}\widehat{\rho}^0\widehat{e}^0=0 \tag{41}$$

Taking Eq. (37) and Eq. (38) into consideration, we can see that $p^0_{\mathrm{i}}=$ Constant, $\forall\,\mathbf{i}$ in space, which ensures that the HLLI method will damp checkerboard modes in the low *Mach* number limit. In addition, Eqs. (39) and (40) indicates that the pressure field obeys a rule: $p(x,t)=p^0(t)+\mathrm{Ma}{\cdot}p^1(x,t)+O(\mathrm{Ma}^2)$. However, Refs [27,28]. have demonstrated the essential features of pressure fluctuations for Euler systems at low speeds:

$$p(x,t)=p^0(t)+M_*^2p^2(x,t) \tag{42}$$

Therefore the HLLI method violates this conclusion and exhibits excessive numerical viscosity at low speeds.

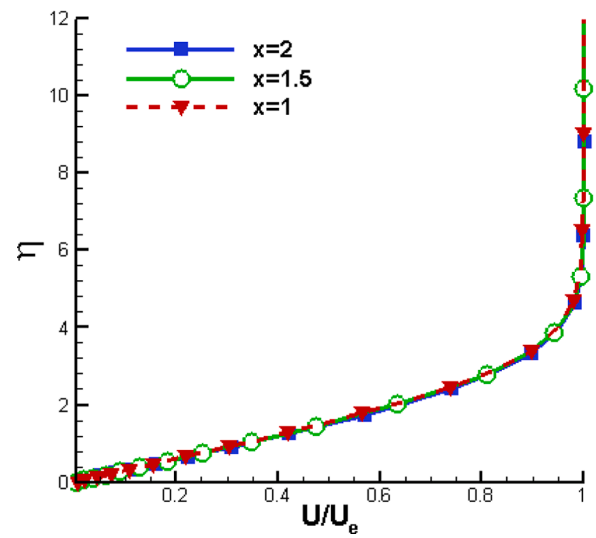


Fig. 7. Normalized velocity profile at different positions computed with HLLI scheme.

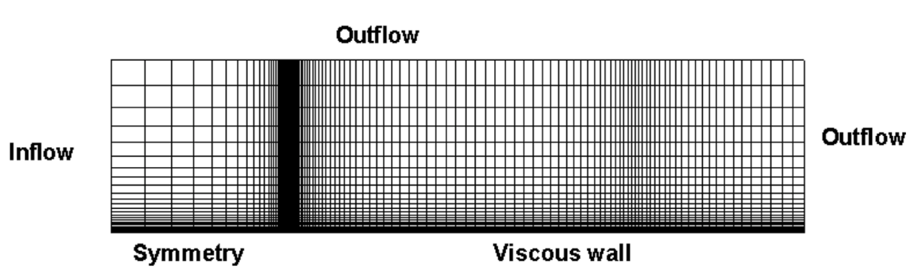


Fig. 6. Computational grid for flat plate boundary layer.

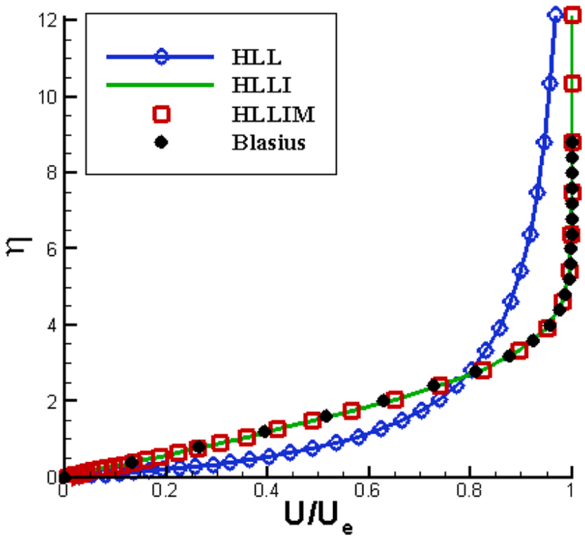


Fig. 8. Comparisons of normalized velocity profile for laminar flow over a flat plate.

Table 1
Computational grids for Odd-even decoupling test.

Computational grid	$\Delta x/\Delta y$(Aspect ratio)
400×20	2
800×20	1
1600×20	0.5

4.3. An improved all-speed Riemann solver HLLIM

4.3.1. Improvement on shock instability

Through the matrix stability analysis, we find the instability may be prohibited if the HLL scheme is implemented in the region of strong shockwaves. Hence we introduce a switch function proposed by Kim et al [6]., which is to control the numerical dissipation when there exists shock waves:

$$\begin{aligned}
&\omega = \min\left(\omega_{p,i-1/2,j}, \omega_{p,i-1/2,j+1}, \omega_{p,i+1/2,j}, \omega_{p,i+1/2,j+1}\right)\\
&\omega_{p,i-1/2,j} = \min\left(p_{i,j}/p_{i-1,j}, p_{i-1,j}/p_{i,j}\right)\\
&\omega_{p,i+1/2,j} = \min\left(p_{i,j}/p_{i+1,j}, p_{i+1,j}/p_{i,j}\right)\\
&\omega_{p,i-1/2,j+1} = \min\left(p_{i,j+1}/p_{i-1,j+1}, p_{i-1,j+1}/p_{i,j+1}\right)\\
&\omega_{p,i+1/2,j+1} = \min\left(p_{i,j+1}/p_{i+1,j+1}, p_{i+1,j+1}/p_{i,j+1}\right)
\end{aligned} \tag{43}$$

The switch function ω is based on a pressure ratio of the surrounding interfaces which suggest that multidimensional effects have been considered about. In the vicinity of strong shockwaves, the switch function approaches to zero and the numerical method is close to the HLL scheme. In other cases, the flux recovers to the HLLI method which is capable of resolving linear degenerate waves. Therefore the resulting numerical flux is written in the following form:

$$\mathbf{f}_{numerical} = \bar{\mathbf{f}}_{HLL} - \omega\cdot\left\{\frac{1}{2}\mathbf{R}_*\left[-\frac{\Delta\xi}{3}(2\delta) + \frac{2\xi_c}{\Delta\xi}\boldsymbol{\Lambda}_*(2\delta)\right]\mathbf{L}_*(\mathbf{q}_R - \mathbf{q}_L)\right\} \tag{44}$$

We also conduct a matrix stability analysis of this scheme and compare the max real part value with the original HLLI scheme Fig. 3. demonstrates that the improved scheme remains stable for a wide range of *Mach* number.

4.3.2. Improvement on low speeds' accuracy

On the viewpoint of physics, Eq. (42) indicates that $\sum_{\mathbf{l}\in\nu(\mathbf{i})} p_{\mathbf{l}}^{1} = 0$ should be satisfied in the low *Mach* number limit. Therefore a modification must be implemented on the right terms of Eqs. (39) and (40) to make them disappear. In this paper, we adopt a scaling function κ in the momentum equations to control the dissipation at low-speeds: $\kappa = \min(M, 1)$. According to the results of the asymptotic analysis in Section 4.2, we construct the formulation of the improved scheme HLLIM:

$$\begin{aligned}
\mathbf{f}_{HLLIM} = &\frac{S_R\mathbf{f}_L - S_L\mathbf{f}_R}{S_R - S_L} + \frac{S_L S_R}{S_R - S_L}\left(\begin{bmatrix}\Delta\rho\\ \kappa\cdot\Delta(\rho u)\\ \kappa\cdot\Delta(\rho v)\\ \Delta(\rho e)\end{bmatrix}\right) - \kappa\cdot\omega\cdot\left\{\frac{1}{2}\mathbf{R}_*\left[\right.\right.\\
&\left.\left. -\frac{\Delta\xi}{3}(2\delta) + \frac{2\xi_c}{\Delta\xi}\boldsymbol{\Lambda}_*(2\delta)\right]\mathbf{L}_*(\mathbf{q}_R - \mathbf{q}_L)\right\}
\end{aligned} \tag{45}$$

For this new formula, we recollect the terms of order 1/Ma of the momentum equations and obtain the following equations:

$$\sum_{\mathbf{l}\in\nu(\mathbf{i})}\left[\hat{p}_{\mathbf{l}}^{1}(n_x)_{\mathbf{il}}\right] = 0 \tag{46}$$

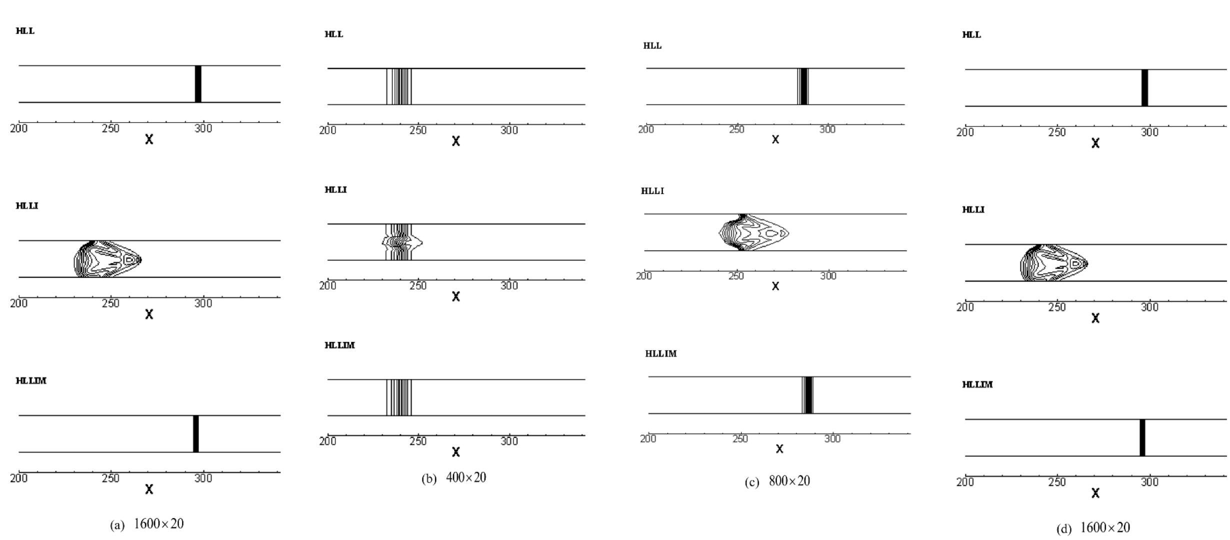
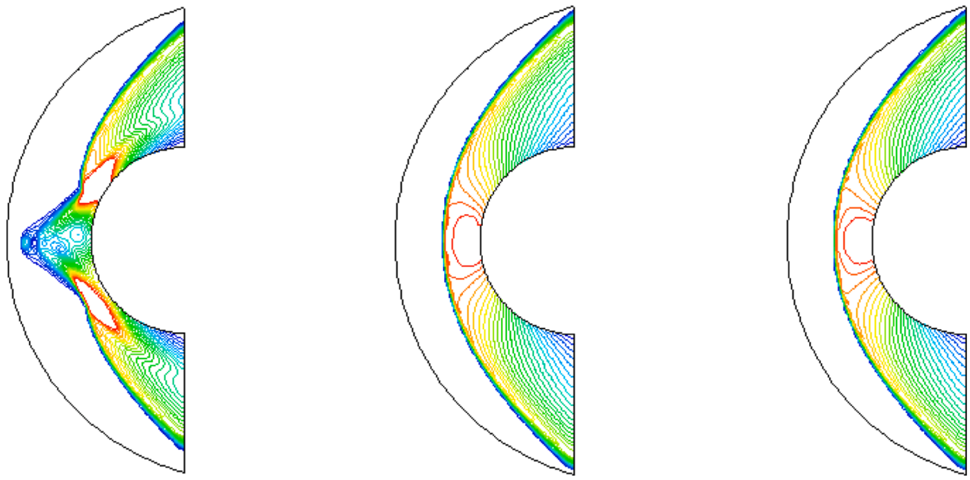
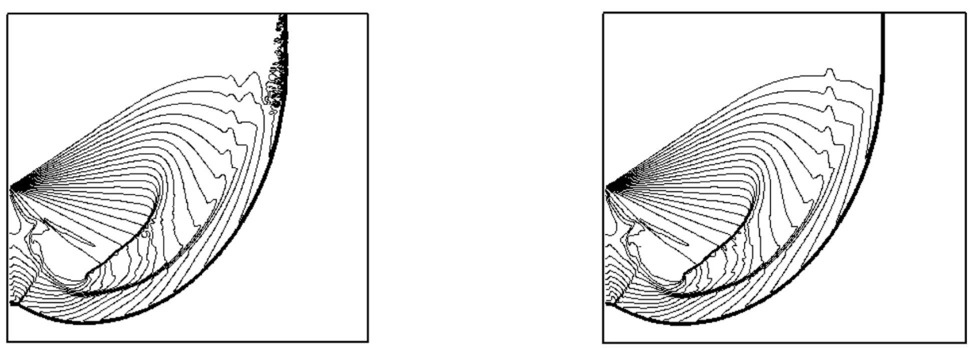


Fig. 9. Density Contours for the Odd-even grid perturbation problem.

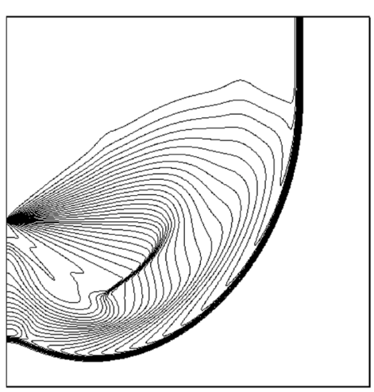


(a)HLLI (b)HLLIM (c)HLL

Fig. 10. Density contours for hypersonic inviscid flow over a cylinder (Ma=20).



(a)HLLI (b)HLLIM



(c)HLL

Fig. 11. Density contours for *Mach* 5.09 normal shock diffraction around a 90° corner.

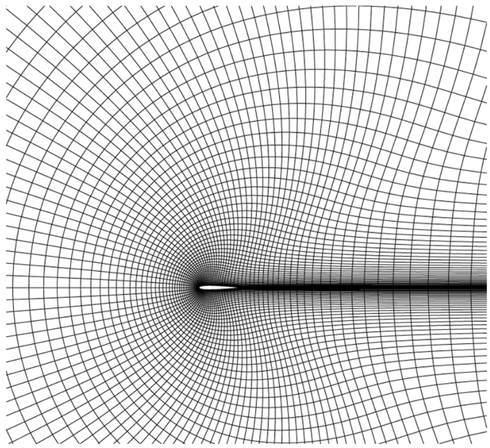


Fig. 12. The primary computational grid of NACA0012 (120 × 50).

$$\sum_{\mathbf{l}\in\nu(\mathbf{i})}\left[\widehat{p}^{1}_{\mathbf{l}}(n_y)_{\mathbf{il}}\right]=0 \tag{47}$$

which implies that $p^1_{\mathbf{i}}=$ Constant, $\forall\,\mathbf{i}$ in space. Therefore the HLLIM scheme satisfies Eq. (42) and is capable of obtaining physical solutions at low speeds theoretically.

5. Numerical experiments

5.1. Linearly degenerate wave resolution

5.1.1. Contact discontinuity

We consider two contact discontinuity problems [29] in this section. The number of the computational grid points is 100 and the initial discontinuity is set in the midpoint of the computational region[0,1]. The first test is a steady contact discontinuity with an initial condition of $(\rho,u,p)_L=(1.4,\ 0,\ 1)$, $(\rho,u,p)_R=(1,\ 0,\ 1)$. And the second contact discontinuity is slowly right moving with an initial condition of $(\rho,u,p)_L=(0.125,0.1125,1),(\rho,u,p)_R=(10,0.1125,1)$. It should be mentioned that the boundaries are zero gradient so that the inner flow field are free from the influences of them. First order reconstruction in spatial and Euler scheme for the time advancing [30] are adopted.

Fig. 4(a) presents the density distribution of the steady contact discontinuity at $t=20$. Here, the HLL scheme is presented as a reference of a non-linear wave degenerate preserving method, and it nearly dissipates this contact discontinuity after a long time run. Hence we demonstrate that the HLLI and the HLLIM scheme are able to capture stationary contact discontinuity exactly, which is consistent with the results of theoretical construction of δ Fig. 4.(b) illustrates density distribution at $t=2$ for the slowly moving contact discontinuity. From this we conclude that HLLIM scheme is capable of capturing contact discontinuity as accurately as HLLI scheme.

5.1.2. Supersonic jet problem

This test is a two-dimensional shear wave problem [31]. The computational region is a square$[0,1]\times[0,1]$ with a 200×200 uniform grid distribution. The initial conditions are set as follows:

$$(\rho,u,v,p)=\begin{cases}0.25,4.0,0.0,0.25 & \mathit{if}\ \ y\ge0.5\\ 1.0,1.4,0.0,0.5 & \mathit{if}\ \ y<0.5\end{cases} \tag{48}$$

The left boundary is kept the same as the upper and lower initial conditions, and other boundaries are set as outflow. Computations are carried out with the same numerical methods in Section 5.1.1 and stopped at $t=5$. All the density residuals have reached a reduction magnitude of 8th orders Fig. 5. gives the density contours, from which we conclude the HLLIM has a similar resolution of HLLI for non grid-aligned shear wave problem.

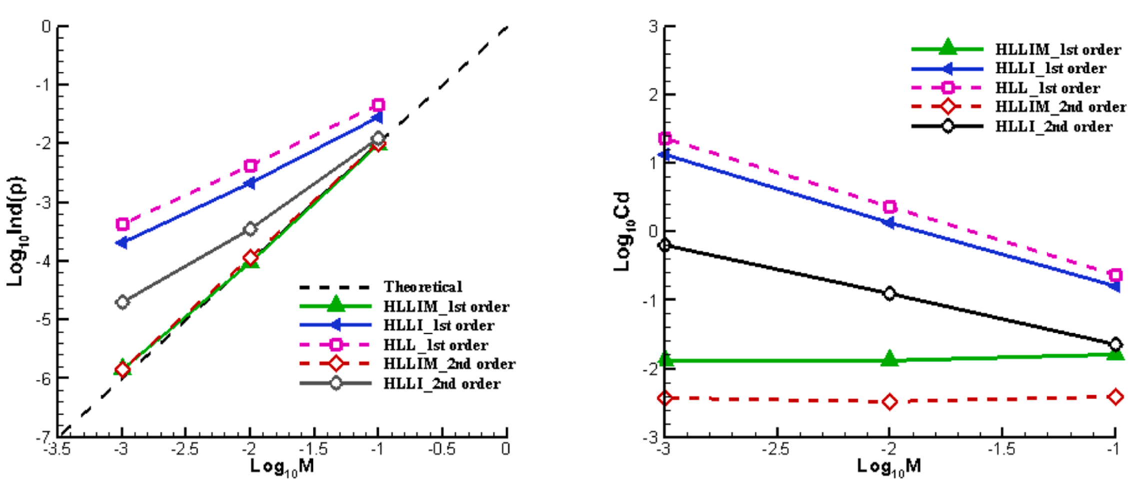
5.1.3. Laminar flow over flat plate

To further demonstrate the linear degenerated waves resolving ability, we perform a laminar flow test over a flat plate [9]. The inflow *Mach* number is 0.3 with a unit Reynolds of 10^5 m^{-1}. Computational grids(120×50) and boundary conditions are illustrated in Fig. 6. Second-order reconstruction is implemented to capture the boundary layer as accurately as possible. For the time advancement, we adopt LU-SGS method with a CFL number of 5 and simulate over 50,000 steps to get convergent results [32]. Results are analyzed with normalized velocity (u/u_∞) and Blasius parameter($\eta=y\sqrt{u_\infty/\mu L}$, where μ is the viscosity coefficient and L is the length from the top of the flat plate) Fig. 7. presents the normalized velocity profile computed by HLLI at different positions, which demonstrates the similarity of the convergent results Fig. 8. compares the results with Blasius' analytical solution. As shown, the HLLIM scheme is capable of resolving the velocity profile as well as HLLI scheme. And both of them are less dissipative than HLL.

5.2. Shock instability problems

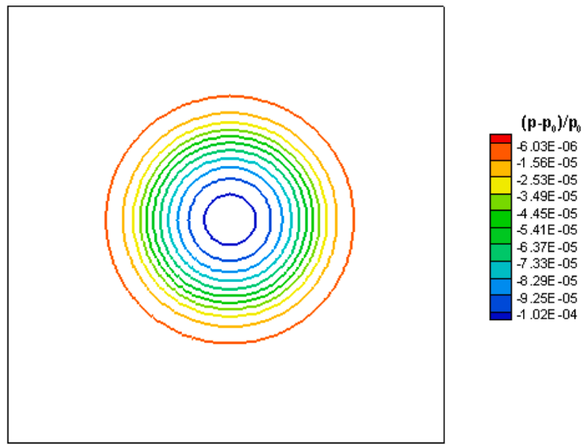
5.2.1. Quirk's test (odd-even decoupling)

Quirk first reported the odd even decoupling problem with grid

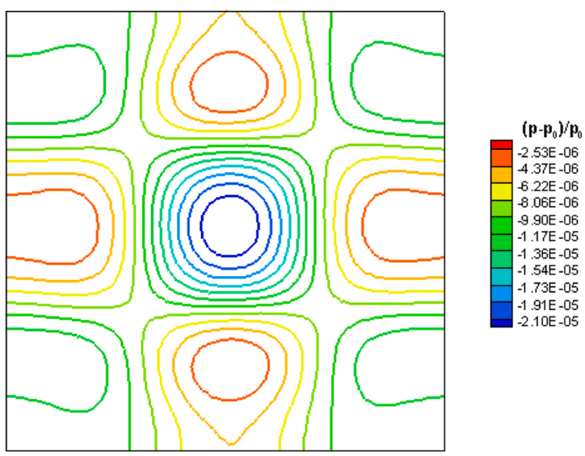


(a) (b)

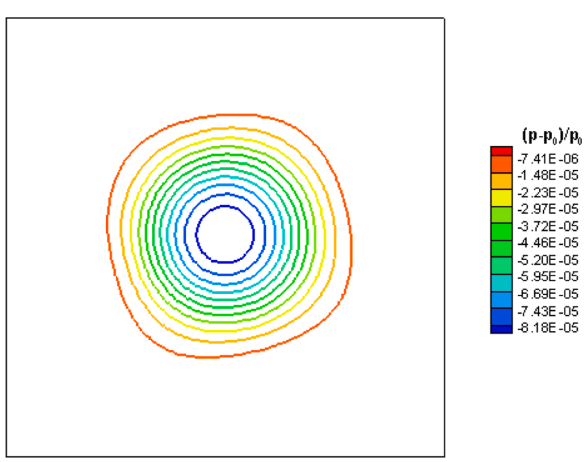
Fig. 13. Inviscid flows over NACA0012 airfoil results. (a)Pressure fluctuations VS Inflow *Mach* number; (b)Drag coefficients VS Inflow *Mach* number.



(a)Initial Condition



(b)HLLI



(c)HLLIM

Fig. 14. Pressure fluctuation($(p - p0) / p0$) contours of a Gresho vortex at Ma=0.01 after one period. (a) Initial condition; (b)HLLI; (c)HLLIM.

perturbations for a grid-aligned moving shock [33]. Odd even decoupling problem perturbs the grids artificially and the grids exhibit a zig-zag pattern in the resolution region. Artificial oscillation is a common phenomenon in numerical calculations and may cause wrong solution when there exists a shock or a contact discontinuity. Here we use this test to study the shock stability performance of the proposed scheme. Researches show that the aspect ratio of cells is a major influencing factor for shock anomaly. Here, three grids with different aspect ratios are investigated and the details are listed in Table 1.

The centerline of the grid is perturbed in the following manner:

$$y_{i,j,mid} = \begin{cases} y_{j,mid} + 10^{-4} for ieven \\ y_{j,mid} - 10^{-4} for iodd \end{cases} \tag{49}$$

A *Mach* 6 shock propagating to right is simulated and pre shock conditions $(\rho, u, v, p) = (1.4, 0, 0, 1)$ are set as initial conditions for the computational domain. The left boundary conditions are set according to post shock values and zero gradient nonreflecting boundary conditions are imposed at the outlet. The top and bottom boundaries are considered as solid walls. For time integration, the explicit 3rd order Runge-Kutta time-marching scheme is used and the CFL number is taken to be 0.5 with the first order reconstruction method for the simulations.

Fig. 9 presents the density contours at a time level of $t = 100$ for different schemes. The HLLI scheme encounters the shock instability, while the HLL scheme and the HLLIM scheme preserved this shock form successfully for different grids. It should be mentioned that positions of the shock are different for different cases. It is because that the time step is different for different grids. Also, in cases the shock anomaly appears, the velocities of the shock are different. However, this will not affect our study of these schemes' robustness.

5.2.2. Hypersonic inviscid flow over a cylinder

As can be seen in the Refs [18,34]., numerous high resolution upwind schemes, such as the Roe scheme [3] and the HLLEM scheme [14], suffer from carbuncle phenomenon for the simulation of hypersonic flows over a blunt body. Here we investigated an inviscid cylinder flow with a *Mach* number of 20. Structured grid with a grid number of $120(circumferential) \times 65(radial)$ is used for the inviscid problem. The first-order reconstruction technology and the implicit LU-SGS scheme are used for the simulations. In addition, all the results are carried out until the density residual achieving fifth-order reduction for this steady problem Fig. 10. shows the density contours of HLLI and HLLIM solvers. As shown, we can find that HLLI solver presents spurious bumps near stagnation streamline while HLLIM solver provides a stable and more accurate solution as the HLL scheme.

5.2.3. Diffraction of a moving normal shock over a corner

This test depicts a *Mach* 5.09 normal shock expanding over a 90° corner [33] which may produce carbuncle problem for some unstable solvers. The computational grid is 400×400 in a unit square $[0,1] \times [0, 1]$, and it is initialized as $(\rho,u,v,p) = (1.4,0,0,1)$. The corner is located at the midpoint of the left boundary. The upper half boundary adopts the post shock conditions while the lower half is treated as a solid wall. The top boundary is also set to wall and the right and bottom boundary are set to zero gradient. Simulations run for $t = 0.1561$ with the same computational methods and CFL number as in Section 5.2.1 Fig. 11. gives the density contours. Spurious oscillations are explicitly observed at the planar shock which clearly demonstrates the instability property of HLLI solver Fig. 11.(b) and (c) present that HLLIM and HLL solver is able to resolve the shock well without shock instability. But HLL solver is with low resolution as it cannot resolve linear discontinuity and is with a higher level of dissipation.

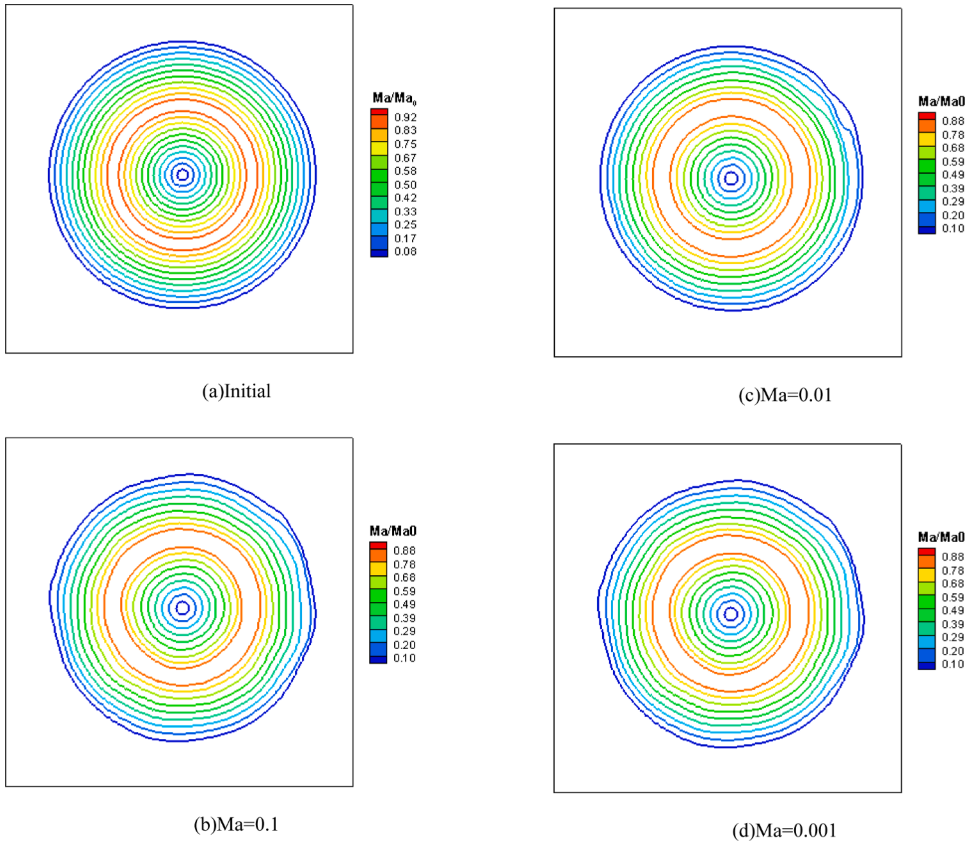


Fig. 15. Computational results of *Mach* ratio isolines with HLLIM solver of different *Mach* numbers after one period. (a)Initial condition; (b)HLLIM solver for Ma=0.1; (c) HLLIM solver for Ma=0.01; (d)HLLIM solver for Ma=0.001.

5.3. Low speed's accuracy (low dissipation property)

5.3.1. 2d inviscid NACA0012 airfoil

We consider the inviscid low-speed flow around NACA0012 airfoil test to assess the low dissipation property of the proposed scheme at low speeds [7,35]. The investigated flow conditions contains Ma=0.001 to 0.1 without angle of attack. The grid is $120(circumferential) \times 50(radial)$ and partially shown in Fig. 12. The implicit LUSGS approach is applied with CFL=5 for over 100,000 steps until the solution achieves convergence. In addition, both the 1st order reconstruction method and the 2nd MUSCL reconstruction method with the van Albada limiter [36] are adopted to ensure that the source of the dissipation is coming from the flux solver.

Fig. 13(a) displays the schemes' pressure fluctuations $\mathrm{Ind}(p) = (P_{\max}\text{-}P_{\min})/P_{\max}$ versus the inlet *Mach* number with different orders of reconstruction methods. The symbol $P_{\max}$ is the largest pressure in the flowfield while the symbol $P_{\min}$ indicates the smallest pressure in the flowfield. According to the theoretical asymptotic predictions, the pressure fluctuations should scale exactly with the square of the *Mach* number in the continuous case with the *Mach* number approaching zero [27]. Therefore, we can find that the HLLIM scheme is with a higher accuracy than the HLLI scheme at low speeds.

Fig. 13(b) shows drag coefficients versus the inlet *Mach* number. Theoretically, the drag of inviscid flow should be zero. Therefore, the computed drag can be a good judgement of numerical error [35]. We can draw a conclusion that the improved HLLIM method is with lower dissipation in the low *Mach* number limit. In addition, as the computational testcases of the HLL scheme with the second order reconstruction method at *Mach* 0.01 and *Mach* 0.001 are diverged, we do not show these results in Fig. 13.

5.3.2. Gresho vortex problem

The problem describes an unsteady flow of a moving vortex proposed by Guillard and Murrone [28]. The computational domain $[0,1] \times [0,1]$ contains a vortex at first with a radius of 0.4 at $(x_0, y_0) = (0.5, 0.5)$. Periodic conditions are set for the left and right boundary, and outflow conditions are implemented for the top and bottom boundary. The initial background state is: $\rho_0 = 1.0$, $p_0 = 1.0$, $\mathbf{u}_0 = (u_0, 0)$. So the sound speed of the background is $a_0 = \sqrt{\gamma \frac{p_0}{\rho_0}} = \sqrt{\gamma}$, and the radius r and moving velocity u_0 is $r = \sqrt{(x - x_0)^2 + (y - y_0)^2}$, and $u_0 = M_0 a_0$, respectively. The initial velocity and pressure are defined as:

$$u_r(r) = u_0 \begin{cases} \dfrac{2r}{R} & if \;\; 0 \le r < \dfrac{R}{2} \\ 2\left(1 - \dfrac{r}{R}\right) & if \;\; \dfrac{R}{2} \le r < R \\ 0 & if \;\; R \le r \end{cases} \tag{50}$$

$$u(x,y) = u_0 - \frac{y - y_0}{r} u_r(x,y) \qquad v(x,y) = \frac{x - x_0}{r} u_r(x,y)$$

$$p(r) = p_0 + u_0^2 \begin{cases} \dfrac{2r^2}{R^2} + 2 - \log 16 & 0 \le r < \dfrac{R}{2} \\ \dfrac{2r^2}{R^2} - \dfrac{8r}{R} + 4\log\left(\dfrac{r}{R}\right) + 6 & if \;\; \dfrac{R}{2} \le r < R \\ 0 & if \;\; R \le r \end{cases} \tag{51}$$

The computational resolution is 100×100 and we computed a period with the explicit 3rd order Runge-Kutta method with a CFL number of 0.8. We consider about an initial *Mach* number of 0.1 to 0.001 and all the simulations are carried out with second-order reconstruction.

Fig. 14 displays the isolines of pressure fluctuation $((p - p_0)/p_0)$

obtained with the HLLI and HLLIM scheme at *Mach* =0.01 after a period time. Initial flowfield is also presented for comparison. The vortex computed with the HLLI scheme causes severe dissipation while the HLLIM scheme preserves the form of the vortex. In addition, we compare the contours of *Mach* number ratio(Ma_{local}/Ma_0) for HLLIM scheme at different initial *Mach* numbers(Fig. 15). The local *Mach* number is computed as [37]:

$$Ma_{local} = \sqrt{\frac{(u-u_0)^2 + v^2}{\gamma p/\rho}} \tag{52}$$

As can be seen, the contours look similar with the same *Mach* ratio number level. Therefore, we can conclude that HLLIM scheme achieves a *Mach* independent accuracy at low speeds for the Gresho vortex test.

6. Concluding remarks

In this work, we review the construction of the HLLI scheme based on the self-similar variable. Then a matrix stability analysis at a wide range of *Mach* numbers and an asymptotic analysis at low speeds are performed to study the properties of the HLLI method. By introducing two switch functions to control the dissipation both in the vicinity of shockwaves and at low speeds, we propose an improved method HLLIM to overcome these unphysical solutions under both circumstances. One and two dimensional test cases demonstrate that the HLLIM is capable of resolving the linear degenerate waves as sharply as the HLLI scheme in Section 5.1. In Section 5.2, a suite of test cases presented that the HLLIM scheme is robust against the shock instability phenomenon while the HLLI scheme cannot avoid this problem. Moreover, the low speeds property has been validated in Section 5.3. Results show that the HLLIM scheme holds the physical pressure fluctuations while the HLL scheme and the HLLI scheme fail to obtain the correct dissipation magnitude. Furthermore, the HLLIM scheme is robust and accurate for all-speed flows. As the HLLI Riemann solver is a universal scheme which is capable of resolving full family of intermediate waves with minimal dissipation for hyperbolic conservation law, analyses in this study could provide reference for the improvement of the HLLI solver for the other hyperbolic conservation law in cases the eigenvalues vary sharply.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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